

EE186 Homework 1

Due: September 29, 2026 (10:00 PM)

1. What's the difference between Cortex-M3 and M4?
2. A microcontroller has 2 KB of byte-addressable SRAM and uses a 12-bit address bus. How many memory locations can the address bus uniquely select? Is the entire SRAM directly addressable with the given bus width? If not, explain why.
3. A 32-bit system stores the 32-bit hexadecimal value 0xDEADBEEF starting at memory address 0x000001B0.
 - (a) Show how the four bytes are stored in memory if the system is little-endian.
 - (b) Show how the four bytes are stored in memory if the system is big-endian.
4. Suppose $r0 = 0x0F0F0F0F$, and $r1 = 0xFEDCBA98$, find the result of the following operations:
 - (a) EOR r3, r1, r0
 - (b) ORR r3, r1, r0
 - (c) AND r3, r1, r0
 - (d) BIC r3, r1, r0
 - (e) BFI r3, r1, #4, #8
 - (f) MVN r3, r1
 - (g) MVN r3, r0
 - (h) MVN r3, r0
ADD r3, r1, r3
5. Suppose $\text{Mask} = 0x00000F0F$ and $P = 0xABCDABCD$. What are the results of the following bitwise operations?
 - (a) $Q = P \ \& \ \text{Mask};$
 - (b) $Q = P \ | \ \text{Mask};$
 - (c) $Q = P \ ^ \ \text{Mask};$
 - (d) $Q = \sim \text{Mask};$
 - (e) $Q = P \ \& \ \sim \text{Mask};$